

STATISTICAL LEARNING THEORY &

CONCENTRATION INEQUALITIES

Logistics

AI2200/EE5603 : Segments 1-2

EE6327 : Segments 1-6

CONCENTRATION :

Toss a fair coin n times. What can we say about fraction of heads?

More generally, if $X_1, X_2, \dots, X_n \sim \text{iid}(p_X)$

what can we say about

$$P_n \left[\left| \frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}X_1 \right| > \epsilon \right] ?$$

$$P_n \left[|f(X_1, \dots, X_n) - \mathbb{E}f(X_i)| > \epsilon \right]$$

Applications

① Statistical learning theory

② Communications

③ Theoretical computer science

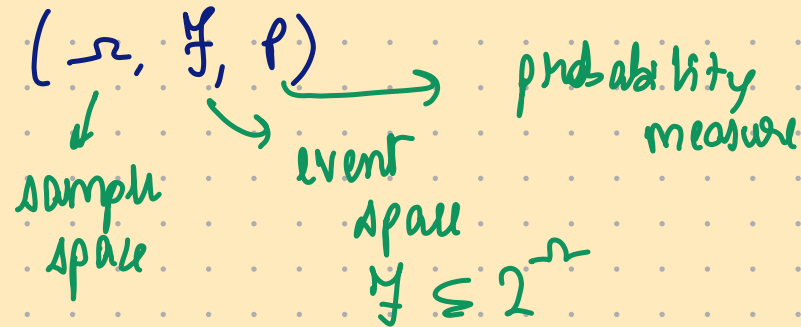
④ Physics

$$y^n = x^n + z^n$$



PROBABILITY 101

Probability space:



① $P(A) \geq 0 \quad \forall A \in \mathcal{F}$

② $P(\Omega) = 1$

③ $A_1, A_2, \dots$ mut. exc.

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

① $\Omega \in \mathcal{F}$

② $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$

③ $A_1, A_2, \dots$ countable $\Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$
 $A_i \in \mathcal{F}$

Q: Why define P on $\mathcal{F}$ & not Ω ?

Q: Why not always take $\mathcal{F} = 2^\Omega$?
(all subsets of Ω)

Vitali set

$$A = [0, 1] = [0, \frac{1}{2}) \cup [\frac{1}{2}, 1] \\ [0, \frac{1}{3}) \cup [\frac{1}{3}, 1]$$

$$A = \bigcup_{i=1}^{\infty} A_i$$

$$\text{Want } \sum_{i=1}^{\infty} P(A) = \sum_{i=1}^{\infty} P(A_i)$$

$$\Omega = \mathbb{R}, \underline{[0, 1]}$$

$$P([0, a])$$

Choose $\mathcal{F}$ to be the smallest σ -algebra that contains all sets of the form $[a, b]$



Borel σ -algebra

Lebesgue σ -algebra

Experiment: Toss wins infinitely many times

$X_1 X_2 X_3 \dots$

HTHTHTHTT
0.101101100...
↑↑↑

$$\mathcal{H} = \left\{ (X_{i_1}, X_{i_2}, \dots, X_{i_k}) \mid i_1 < i_2 < \dots < i_k, \forall k \right\}$$

$$X = \sum_{i=1}^{\infty} x_i 2^{-i} = g(X_1, X_2, \dots)$$

$$\{(x_1, x_2, \dots) : g(x_1, \dots) \leq \alpha\}$$

$$\{X < \frac{1}{2}\} \equiv \{\text{seq. st } X_0 = 0\}$$

$$\{X < \frac{1}{4}\} \equiv \left\{ \begin{array}{c} \text{---} \text{---} \\ X_0 = 0 \\ X_1 = 0 \end{array} \right\}$$

Random variable $X: \Omega \rightarrow \mathbb{R}$

$$\forall \alpha \in \mathbb{R}, \quad X^{-1}(\underbrace{(-\infty, \alpha]}_{X \leq \alpha}) \in \mathcal{F}$$

Notation :

x	deterministic value
X	random variable
$\underline{x}, x^n$	deterministic vector
$\underline{X}, X^n$	random vector

Probability measure for random variables

$$P(X \in (-\infty, \alpha]) = F_X(\alpha)$$

$$f_X(\alpha) = \frac{d}{d\alpha} F_X(\alpha)$$

$$p_X(\alpha) = \Pr[X = \alpha]$$

Moments

Non central : $\mathbb{E}[X^k]$

Central : $\mathbb{E}[(X - \mathbb{E}X)^k]$

Moment generating function :

$$\phi_X(t) = \mathbb{E} e^{tX}$$

Q: How do you obtain $\mathbb{E}X^k$ from $\phi_X(t)$?

Common distributions: Bernoulli, Binomial, Poisson, Geometric, Uniform, Exponential,
Gaussian, Chi squared, Gamma

Jensen's inequality: $f: \mathbb{R} \rightarrow \mathbb{R}$

f is convex if

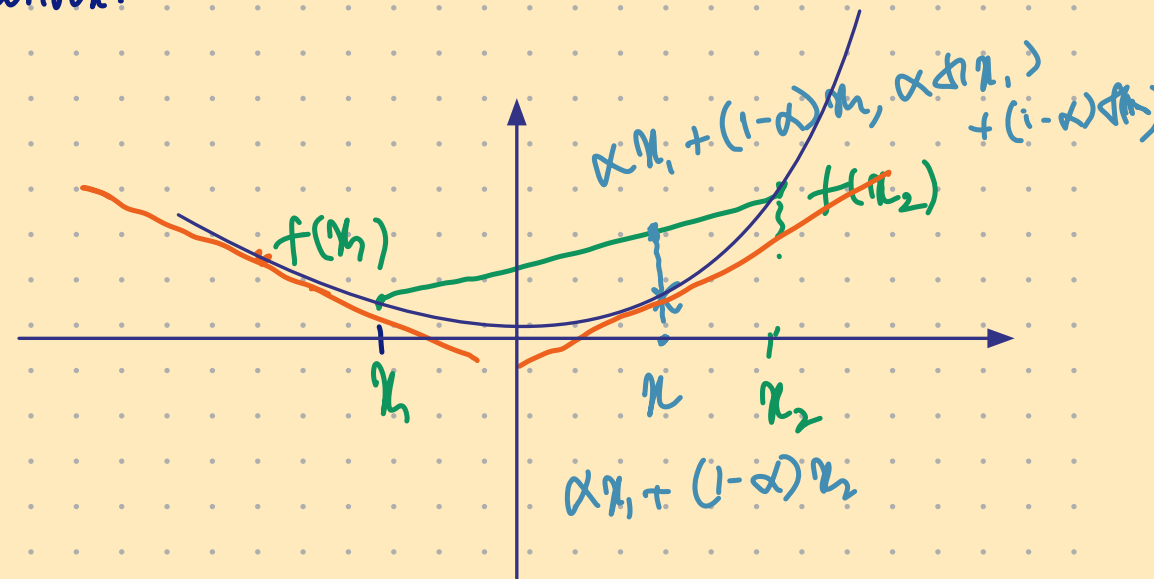
$$f(\alpha x_1 + (1-\alpha)x_2) \leq \alpha f(x_1) + (1-\alpha)f(x_2)$$

$$\forall x_1, x_2 \in \mathbb{R}$$

$$\forall 0 \leq \alpha \leq 1$$

$\equiv$ f is convex
if $f''(x) \geq 0$

f is concave if $-f$ is convex.



Theorem: If f is convex & X a r.v.,

$$f(E(X)) \leq E(f(X))$$

Basic tail bounds

Markov Inequality

Let X be a n.v. & $\mathbb{E}X = \mu < \infty$
 $X \geq 0$ w.p.1

$$\forall \delta > 0 \quad \Pr[X \geq \delta] \leq \frac{\mu}{\delta}$$

Proof:

$$\begin{aligned} \mu &= \int_0^{\infty} x f_X(x) dx \\ &= \int_0^{\delta} x f_X(x) dx + \int_{\delta}^{\infty} x f_X(x) dx \\ &\geq \underbrace{0}_{\geq 0} + \int_{\delta}^{\infty} \delta f_X(x) dx \\ &= \delta \Pr[X \geq \delta] \end{aligned}$$

$$X = 1 \text{ w.p. } 1$$

$$\mathbb{E} X = 1$$

$$\sigma = 1$$

$$\Pr[X \geq 1] = 1$$

Chebyshev inequality

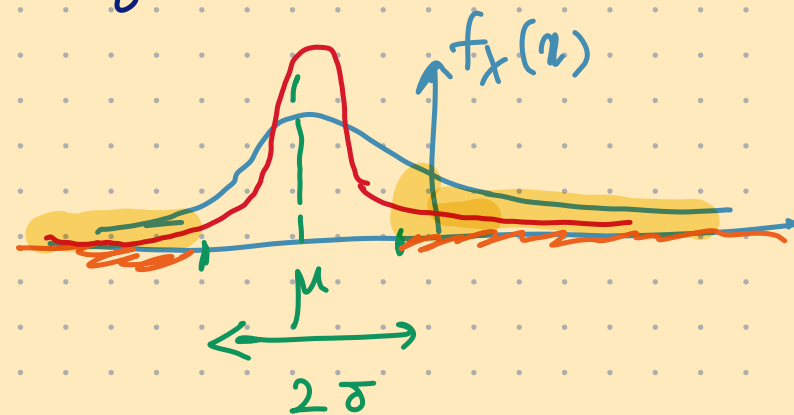
Theorem : X is a RV with $\mathbb{E}X = \mu < \infty$
 $\text{Var}(X) = \sigma^2 < \infty$

Then, $\forall \delta > 0$,

$$\mathbb{P}n[|X - \mu| \geq \delta] \leq \frac{\sigma^2}{\delta^2}$$

Proof :

$$\begin{aligned} \mathbb{P}n[|X - \mu| \geq \delta] &= \mathbb{P}n[(X - \mu)^2 \geq \delta^2] \\ &= \mathbb{P}n[Y \geq \delta^2] \\ &\leq \frac{\mathbb{E}Y}{\delta^2} = \frac{\mathbb{E}(X - \mu)^2}{\delta^2} = \frac{\sigma^2}{\delta^2} \end{aligned}$$



$$X_1, X_2, \dots, X_N \sim \text{iid} \quad \mathbb{E}X = \mu$$

$$A_N = \frac{1}{N} \sum_{i=1}^N X_i \xrightarrow{P} \mu$$

$$\text{WLLN: } \forall \epsilon > 0 \quad \Pr[|A_N - \mu| > \epsilon] \rightarrow 0 \quad \text{as } N \rightarrow \infty$$

$$\begin{aligned} \mathbb{E} A_N &= \mu & \text{Var}(A_N) &= \text{Var}\left(\frac{1}{N} \sum_{i=1}^N X_i\right) \\ & & &= \frac{1}{N^2} \sum_{i=1}^N \text{Var}(X_i) = \frac{\sigma^2}{N} \end{aligned}$$

$$\Pr[|A_N - \mu| \geq \epsilon] \leq \frac{\sigma^2}{N \epsilon^2} \rightarrow 0 \quad \text{as } N \rightarrow \infty$$

① May not be the best rate of convergence
(for specific f_X)

② WLLN holds even if σ^2 is not finite

Generalization :

$$P_n[|X - \mu| \geq \delta] \leq \frac{E(X - \mu)^{2k}}{\delta^{2k}}$$

$$P_n[|A_n - \mu| \geq \delta] \leq O\left(\frac{1}{N^{2k-1}}\right)$$

CHERNOFF - CRAMER BOUNDS

Theorem: X is a rv with MGF ϕ_X

Then, $\forall \delta > 0$,

Right tail $Pn[X - \mu \geq \delta] \leq \frac{\phi_X(t)}{e^{t(\delta + \mu)}} \quad \forall t > 0$

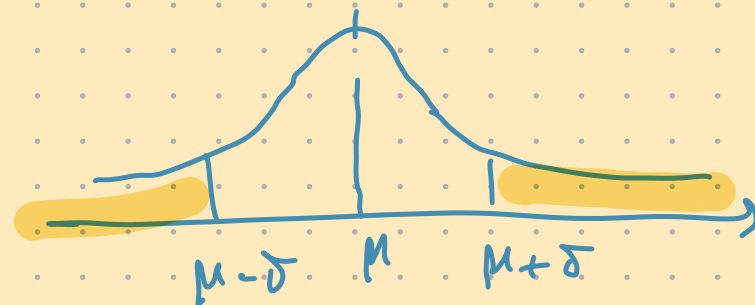
Left tail $Pn[X - \mu \leq -\delta] \leq \frac{\phi_X(-t)}{e^{t(\delta - \mu)}} \quad \forall t > 0$

Proof: $Pn[X - \mu \geq \delta]$

$$= Pn[t(X - \mu) \geq t\delta] \quad \forall t > 0$$

$$= Pn[e^{t(X - \mu)} \geq e^{t\delta}]$$

$$= Pn[e^{tX} \geq e^{t(\mu + \delta)}]$$



as exp is increasing

$$\leq \frac{\mathbb{E} e^{tX}}{e^{t(\mu+\sigma)}} \quad (\text{Markov})$$

$$= \frac{\phi_X(t)}{e^{t(\mu+\sigma)}}$$

$$A_N = \frac{1}{N} \sum_{i=1}^N X_i$$

$$S_N = \sum_{i=1}^N X_i$$

$$\begin{aligned} \phi_{A_N}(t) &= \mathbb{E} e^{tA_N} = \mathbb{E} e^{\frac{t}{N} \sum_{i=1}^N X_i} \\ &= \mathbb{E} \left[\prod_{i=1}^N \underbrace{e^{\frac{t}{N} X_i}}_{Y_i} \right] \\ &= \prod_{i=1}^N \mathbb{E} \left[e^{\frac{t}{N} X_i} \right] \\ &= \left(\mathbb{E} e^{tX/N} \right)^N = \left(\phi_X\left(\frac{t}{N}\right) \right)^N \end{aligned}$$

$$P_n[A_n \geq \mu + \delta] \leq \frac{\left(\phi_x\left(\frac{t}{N}\right)\right)^N}{e^{\frac{t(\mu + \delta)}{N}}} \quad \forall t > 0$$

$$P_n[A_n \leq \mu - \delta] \leq \frac{\left(\phi_x\left(-\frac{t}{N}\right)\right)^N}{e^{\frac{t(\delta - \mu)}{N}}}$$

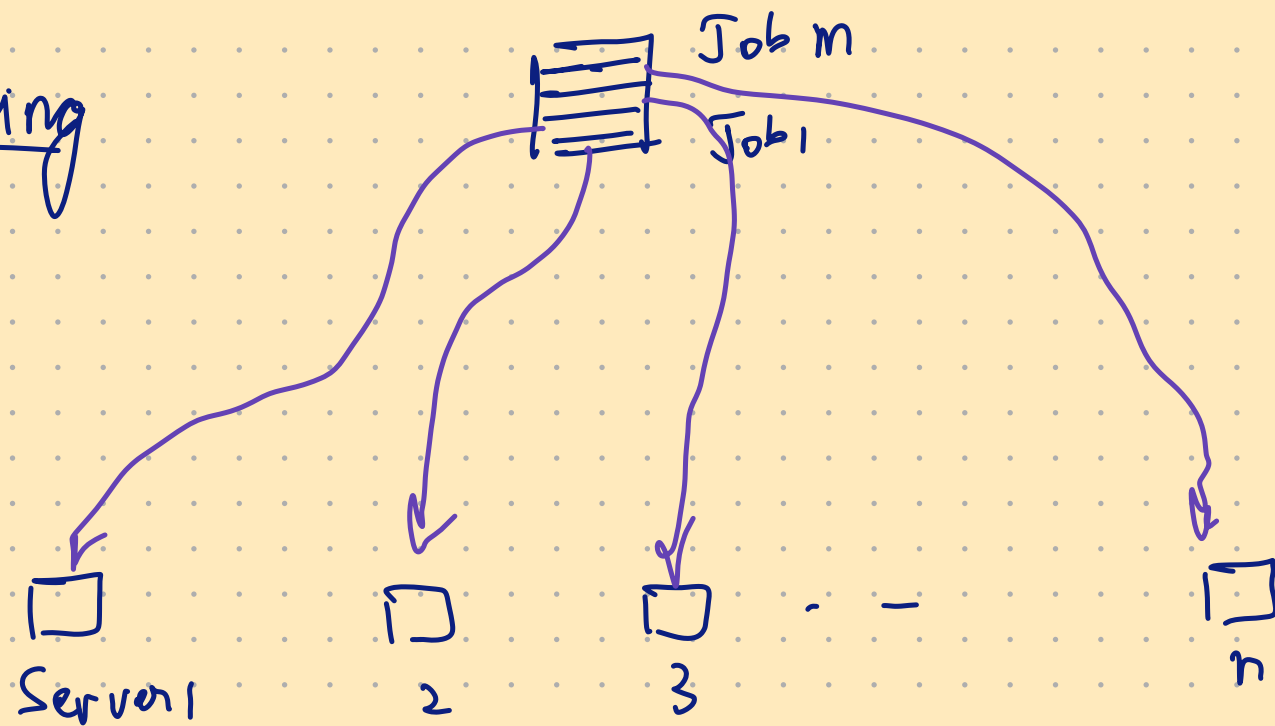
Thm: If $X_1, X_2, \dots, X_n$ are iid $\text{Ber}(p)$ r.v.

Then

$$P_n[|A_n - p| \geq \delta p] \leq \begin{cases} 2 e^{-N \delta^2 p / 3} & 0 < \delta < 1 \\ 2 e^{-N \delta^2 p / (2 + \delta)} & \delta > 0 \end{cases}$$

$$P_n[A_n \geq p(1 + \delta)]$$

load balancing



Random allocation: Allocates to a server chosen uniformly at random

Let $x_{ij} = \begin{cases} 1 & \text{if } i^{\text{th}} \text{ job is allotted to server } j \\ 0 & \text{else} \end{cases}$

$X_{i1} X_{i2} \dots X_{in}$ Not iid

$X_{1j} X_{2j} \dots X_{mj}$ iid

$$X = \begin{matrix} & \begin{matrix} \text{server} \\ \uparrow \\ n \end{matrix} \\ \begin{matrix} m \\ \downarrow \\ j \end{matrix} & \left[\begin{matrix} \\ \\ \end{matrix} \right] \end{matrix}$$

$$X_{ij} \sim \text{Ber}\left(\frac{1}{n}\right)$$

Load for server j : $L_j = \sum_{i=1}^m X_{ij}$

$$\mathbb{E} L_j = \frac{m}{n}$$

$$\begin{aligned} & \Pr\left[L_j - \frac{m}{n} \geq \frac{m\delta}{n}\right] \\ &= \Pr\left[\frac{L_j}{m} - \frac{1}{n} \geq \frac{\delta}{n}\right] \leq 2 \exp\left[-\frac{m\delta^2/n}{2+\delta}\right] \end{aligned}$$

$$Pr \left[L_j \geq \frac{m}{n} + \frac{m\delta}{n} \text{ for some } j \right]$$

$$= Pr \left[\bigcup_{j=1}^n \left\{ L_j \geq \frac{m}{n} + \frac{m\delta}{n} \right\} \right]$$

$$\leq \sum_{j=1}^n Pr \left[L_j \geq \frac{m}{n} + \frac{m\delta}{n} \right]$$

$$\leq n \cdot 2 \exp \left[-\frac{m\delta^2}{n(2+\delta)} \right]$$

$$\leq \theta\left(\frac{1}{n}\right) \quad \text{if} \quad \delta = \theta(\log n)$$

$$\text{If } m = n, \quad \text{Max-load} \leq 1 + \theta(\log n)$$

$\swarrow$ w.h.p.
 $\log \log n$

Measure of performance: $X \rightarrow g(X)$

Goal: Design g s.t. $\Pr[g(X) \neq Y]$ is small

Only given S

$$g^* = \arg \min_{g \in \mathcal{H}} \Pr[g(X) \neq Y]$$

PAC (Probably Approximately Correct) learning:

$$R(g) = \Pr[g(X) \neq Y] \quad \text{risk achieved by } g$$

Learning algorithm: Rule that takes S as i/p
o/p: $\hat{g}_S$

Empirical risk
Minimization
(ERM)

$$\hat{g}_s =$$

argmin
g ∈ G

$$\frac{1}{n} \sum_{i=1}^n$$

$$1_{\{g(x_i) \neq y_i\}}$$

$$= \begin{cases} 1 & \text{if } g(x_i) \neq y_i \\ 0 & \text{else} \end{cases}$$

$$\Pr[R(\hat{g}) \geq R(g^*) + \epsilon] \leq \delta$$

↓
evaluated
wrt dataset S

↓
evaluated wrt a new sample indep of S

We say that $\mathcal{Y}$ (hypothesis class) is PAC learnable
if $\forall \epsilon, \delta > 0, \exists N_{\epsilon, \delta, \mathcal{Y}}$ st $\exists$ a learning algo
that takes as i/p N iid samples $S = \{(X_1, Y_1), \dots, (X_N, Y_N)\}$
& o/p $\hat{g} \in \mathcal{Y}$ st
$$\Pr[R(\hat{g}) \geq R(g^*) + \epsilon] \leq \delta$$

as long as $N \geq N_{\epsilon, \delta, \mathcal{Y}}$

* Assume that $\mathcal{Y}$ is finite

* How well does ERM perform?

$$R(g) = \Pr[f(x) \neq y] \rightarrow \text{number that dep on } f, x, y$$

Empirical risk

$$R_S(g) = \frac{1}{N} \sum_{i=1}^N \mathbb{1}_{\{g(x_i) \neq y_i\}} \rightarrow \text{random qty depends on } S$$

$$\mathbb{E} R_S(g) = R(g)$$

iid $\text{Bern}(R(g))$ r.v.

For any $g \in \mathcal{G}$,

$$\Pr[|R_S(g) - R(g)| \geq \epsilon] \leq \Pr[|R_S(g) - R(g)| \geq \epsilon R(g)]$$

(for small ϵ)

$$\leq 2 \exp\left[-\frac{N \epsilon^2 R(g)}{3}\right]$$

Assume $R(g) \geq m = R(g^*)$

$$\leq 2 \exp\left[-\frac{N \epsilon^2 R(g^*)}{3}\right]$$

$$\Pr[R(\hat{g}_S) \geq R(g^*) + \epsilon]$$

↓
ERM

$$\Pr[|R_S(\hat{g}_S) - \underbrace{R(\hat{g}_S)}_{\text{depends on } S}| \geq \epsilon]$$

$$\Pr[|R_S(g) - R(g)| \geq \epsilon \text{ for some } g \in \mathcal{Y}]$$

$$= \Pr[\bigcup_{g \in \mathcal{Y}} \{ |R_S(g) - R(g)| \geq \epsilon \}]$$

$$\leq \sum_{g \in \mathcal{Y}} \Pr[|R_S(g) - R(g)| \geq \epsilon]$$

$$\leq |\mathcal{Y}| \exp\left[-\frac{N \epsilon^2 R(g^*)}{3} \right]$$

Suppose $|R_S(q) - R(q)| \leq \epsilon \quad \forall q \in Y$

$$\Rightarrow |R_S(\hat{q}_S) - R(\hat{q}_S)| \leq \epsilon$$

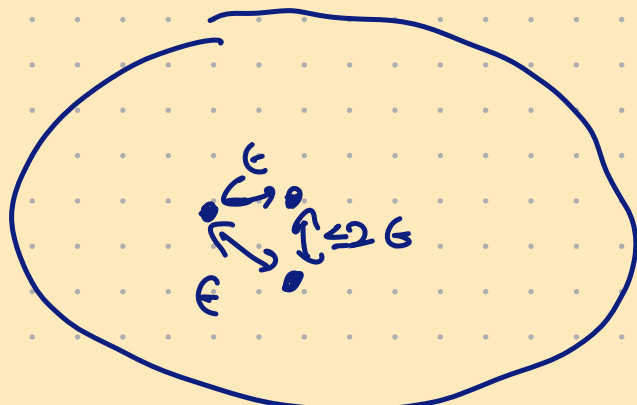
$$R(\hat{q}_S) \leq R_S(\hat{q}_S) + \epsilon \quad \text{--- (1)}$$

$$|R_S(q^*) - R(q^*)| \leq \epsilon$$

$$\Rightarrow R_S(q^*) \leq R(q^*) + \epsilon \quad \text{--- (2)}$$

By defn $R_S(\hat{q}_S) \leq R_S(q^*) \quad \text{--- (3)}$

$$R(\hat{q}_S) \leq R_S(\hat{q}_S) + \epsilon \leq R_S(q^*) + \epsilon + \epsilon \\ = R(q^*) + 2\epsilon$$



Lemma : If $|R_S(g) - R(g)| \leq \epsilon \quad \forall g \in \mathcal{Y}$

$$\text{Then, } R(\hat{g}_S) - R(g^*) \leq 2\epsilon$$

$$P_n \left[|R_S(\hat{g}_S) - R(\hat{g}_S)| \geq \epsilon \right] \leq |\mathcal{Y}| 2 \exp \left[-\frac{N \epsilon^2 R(g^*)}{3} \right]$$

From Lemma

$$P_n \left[R(\hat{g}_S) \geq R(g^*) + \epsilon \right] \leq |\mathcal{Y}| 2 \exp \left[-\frac{N \epsilon^2 R(g^*)}{3} \right]$$
$$|\mathcal{Y}| 2 \exp \left[-\frac{N \epsilon^2 R(g^*)}{3} \right] \leq \delta$$

It

$$N \geq \frac{3 \log 2 |\mathcal{Y}|}{\epsilon^2 R(g^*) \delta}$$

$$P_n \left[R(\hat{g}) \geq R(g^*) + \epsilon \right] \leq \delta$$

Q: How do concentration inequalities help?

Bound: $\Pr\left[\left|\frac{1}{N} \sum_{i=1}^N X_i - \mu\right| \geq \epsilon\right]$

Approach 1: Find distribution of $\frac{1}{N} \sum_{i=1}^N X_i$

Approach 2: Use Chernoff-type bounds for
avg of iid rv

(Easier than finding distribution)

Many cases, approach 2 is good enough.

More general, $\Pr[f(X_1, \dots, X_n) - \mu] \geq \epsilon$

Eg: $X_1, \dots, X_n \sim \text{iid } P_X$

$f(X_1, \dots, X_n) = \# \text{ of distinct values in } X_1, \dots, X_n$

Q: Benefit of deriving bounds for PAC learning

Approach 1: For given learning task, try different algorithms & see what works best

- Is there a more systematic approach?
- How do we know if we can do better?

SLT approach:

- Derive fundamental limits on the performance of any learning algorithm
- Design fundamental techniques for achieving these bounds

Example where $R_S(\hat{g}_S)$ does not concentrate.

$$\mathcal{X} = \mathbb{R}, \quad \mathcal{Y} = \{0, 1\}$$

$$g = \left\{ g_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases} : A \in \mathbb{R} \right\}$$

For any fixed A , & $X_1, \dots, X_N \sim \text{iid } p_{X,Y}$,

$$R(g_A) = \Pr[g_A(X) \neq Y]$$

$$R_S(g_A) = \frac{1}{N} \sum_{i=1}^N \mathbb{1}_{\{g_A(X_i) \neq Y_i\}}$$

$$\mathbb{E} R_S(g_A) = R(g_A)$$

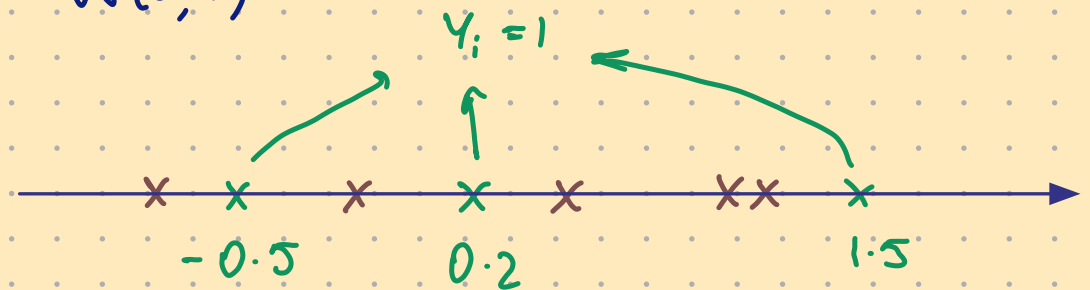
$$\Pr[|R_S(g_A) - R(g_A)| \geq \epsilon] \leq 2^{-cN\epsilon^2}$$

$$S = \{(X_1, Y_1) \dots (X_N, Y_N)\} \sim \text{iid } (p_{XY})$$

$$\hat{g}_S = \underset{g \in \mathcal{Y}}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^N \mathbb{1}_{g(X_i) \neq Y_i}$$

Consider $A = \{X_1 \dots X_N\}$ st $Y_i = 1$

Assume $p_X \sim \mathcal{N}(0, 1)$



$$A = \{-0.5, 0.2, 1.5\}$$

$$R_S(g_A) = 0$$

$$\text{But } R(g_A) = \Pr_{(X, Y)} [g_A(X) \neq Y] = \Pr[Y = 1]$$

Ex Random graph on n vertices. Vertex i & j
are independently connected w.p. p
(Erdős - Rényi)

Find $\mathbb{E}[\# \text{ edges}]$

$\mathbb{E}[\text{degree of } i^{\text{th}} \text{ vertex}]$
 D_i

Lower bound on $\Pr[|D_i - \mathbb{E}D_i| \leq \delta \mathbb{E}D_i, \text{ for all } i]$

S.T. if $p = \frac{1}{n}$,

$\Pr[\exists \text{ at least one } i \text{ s.t. } D_i = \Omega\left(\frac{\log n}{\log \log n}\right)] \geq 0.9$

Ex Let $X_1, \dots, X_N$ i.i.d. Rademacher r.v. & $\underline{a} \in \mathbb{R}^N$

$$x_i \in \{1, -1\}$$

$$P_{X_i}(1) = P_{X_i}(-1) = \frac{1}{2}$$

Show that $P_n \left[\left| \sum_{i=1}^N a_i X_i \right| \geq \delta \right] \leq 2 \exp \left[-\frac{\delta^2 N}{2 \|\underline{a}\|^2} \right]$

(Hint: Use $\frac{e^x + e^{-x}}{2} \leq e^{x^2/2}$)

Ex: $P_n \left[A_N - \mu \geq \delta \right] \leq \left(\frac{\phi_X(t)}{e^{t(\delta + \mu)}} \right)^N$

$(X_1, \dots, X_N \text{ i.i.d.})$

$P_n \left[A_N - \mu \leq -\delta \right] \leq \left(\frac{\phi_X(-t)}{e^{t(\delta - \mu)}} \right)^N$

$\downarrow$

$\frac{1}{N} \sum_{i=1}^N X_i$

Chernoff bound for Gaussian

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$\phi_X(t) = \exp\left[\mu t + \frac{\sigma^2 t^2}{2}\right]$$

$$P_n[X - \mu \geq \delta] \leq \frac{\phi_X(t)}{\exp[t(\mu + \delta)]}$$

$$= \exp\left[\cancel{\mu t} + \frac{\sigma^2 t^2}{2} - \cancel{\mu t} - t\delta\right]$$

$$= \exp\left[-t\delta + \frac{\sigma^2 t^2}{2}\right]$$

$$\forall t > 0$$

$$-\delta + \sigma^2 t = 0$$

$$t = \frac{\delta}{\sigma^2}$$

$$P_n[X - \mu \geq \sigma] \leq \exp\left[-\frac{\sigma^2}{2} + \frac{\cancel{\sigma^2}}{2} \frac{\sigma^2}{\cancel{\sigma^2}}\right]$$

$$= \exp\left[-\frac{\sigma^2}{2\sigma^2}\right]$$

$$P_n[X - \mu \leq -\sigma] \leq \exp\left[-\frac{\sigma^2}{2\sigma^2}\right]$$

For any X $\mathbb{E}X = \mu$

$$P\left[\left|\frac{1}{N} \sum_i X_i - \mu\right| > \epsilon\right] \rightarrow 0$$

as $N \rightarrow \infty$

Ex

Assume $X \sim N(0, 1)$

$$\text{ST} \quad \mathbb{E}|X|^p = \frac{2^{p/2} \Gamma\left(\frac{p+1}{2}\right)}{\Gamma(1/2)}$$

$$\therefore (\mathbb{E}|X|^p)^{1/p} = o(\sqrt{p}) \quad \text{as } p \rightarrow \infty$$

$$\text{where:} \quad \Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$$

$$\text{Stirling's formula:} \quad \Gamma(n+1) = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n (1+o(1))$$

Assume $X_1, \dots, X_n \sim \text{iid } \mathcal{N}(0, 1)$ & $\underline{a} \in \mathbb{R}^n$

$$Y = \sum_{i=1}^n a_i X_i$$

THEOREM

Let X be a random variable

The following are equivalent (here, $\frac{K_i}{K_j}$ can be chosen absolute constant that does not depend on f_X)

$$\textcircled{1} \quad \Pr[|X| \geq \delta] \leq 2 \exp\left[-\frac{\delta^2}{K_1^2}\right] \quad \forall \delta \geq 0$$

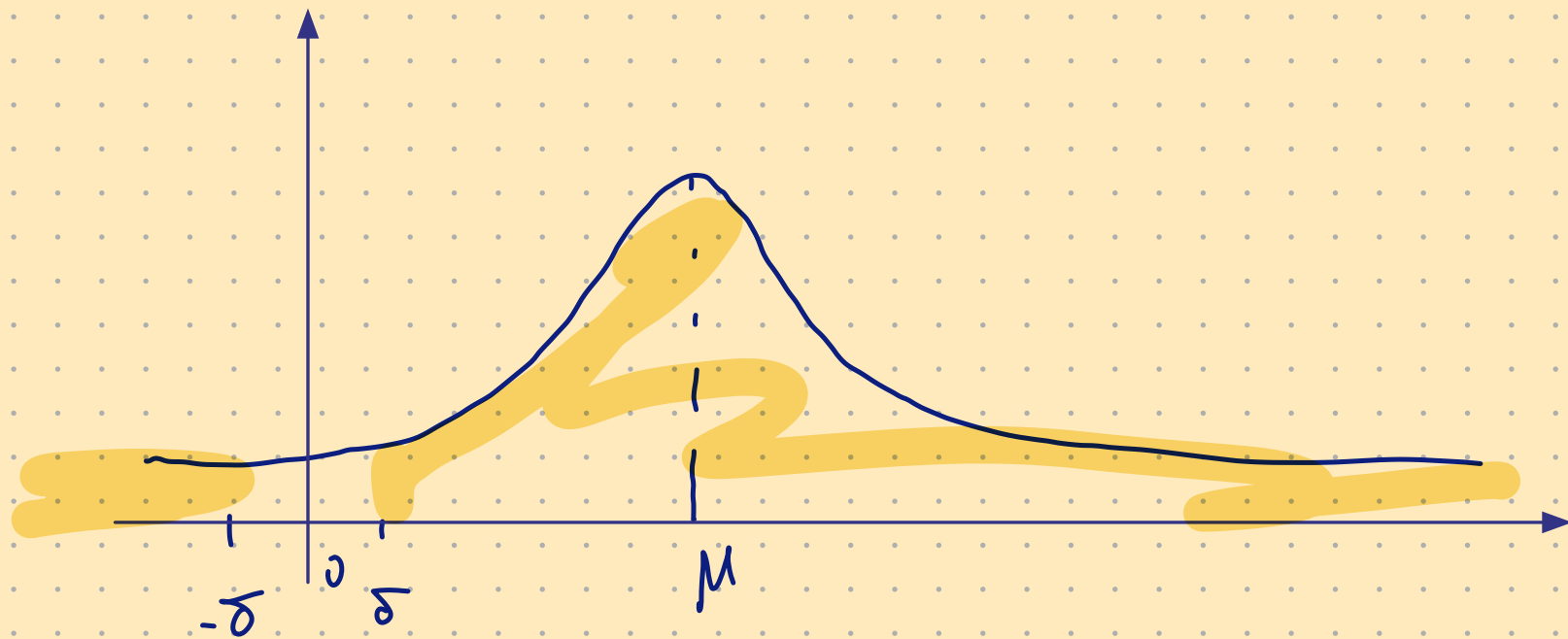
$$\textcircled{2} \quad \left(\mathbb{E}|X|^p\right)^{1/p} \leq K_2 \sqrt{p} \quad p \geq 1$$

$$\textcircled{3} \quad \mathbb{E} e^{t^2 X^2} \leq e^{K_3^2 t^2} \quad \forall t: |t| \leq \frac{1}{K_3}$$

$$\textcircled{4} \quad \mathbb{E} e^{X^2/K_4^2} \leq 2$$

If $\mathbb{E}X = 0$,

$$\textcircled{5} \quad \mathbb{E} e^{tX} \leq e^{K_5^2 t^2} \quad \forall t \in \mathbb{R}$$



R.V that is not subgaussian:

$$f_X(x) = \frac{e^{-\beta x}}{\beta}, \quad x > 0$$

Subgaussian random variable: Any x that satisfies the above

Subgaussian norm:

$$\|X\|_{\psi_2} = \argmin \{ K > 0 : \mathbb{E} e^{X^2/K^2} \leq 2 \}$$

Basic observations:

① If x is SG, then αX is SG

$$\begin{array}{ccc} \|\alpha X\|_{\psi_2} & = & |\alpha| \|X\|_{\psi_2} \\ \downarrow & & \downarrow \\ Y & & K \end{array}$$

$$\mathbb{E} e^{y^2/\alpha^2 k^2} = \mathbb{E} e^{x^2/k^2} \leq 2$$

② Take $X \sim \mathcal{N}(0, \sigma^2)$

$$\phi_{X^2}(t) = \frac{1}{(1 - 2\sigma^2 t)^{1/2}} \quad t < \frac{1}{2\sigma^2}$$

$$\mathbb{E} e^{X^2/k^2} \leq 2$$

Smaller k st $\phi_{X^2}\left(\frac{1}{k^2}\right) \leq 2$

Larger t st $\phi_{X^2}(t) \leq 2$

$$\frac{1}{\sqrt{1 - 2\sigma^2 t}} = 2$$

$$1 - 2\sigma^2 t = \frac{1}{4}$$

$$\frac{3}{4} = 2\sigma^2 t$$

$$t = \frac{3}{8\sigma^2}$$

$$K^2 = \frac{8\sigma^2}{3} = \|X\|_{\psi^2}^2$$

$f: \mathcal{H} \rightarrow \mathbb{R}_{\geq 0}$ is a norm if

$$\textcircled{1} \quad \begin{aligned} f(x) &\geq 0 \quad \forall x \in \mathcal{H} \\ &= 0 \quad \text{iff } x = 0 \end{aligned}$$

$$\textcircled{2} \quad f(\alpha x) = |\alpha| f(x) \quad \forall \alpha \in \mathbb{R}$$

$$\textcircled{3} \quad \begin{aligned} f(x+y) &\leq f(x) + f(y) \\ &\quad \forall x, y \in \mathcal{H} \end{aligned}$$

$$\|X\|_{\psi_2} = \inf \left\{ K > 0 : \mathbb{E} e^{X^2/K^2} \leq 2 \right\}$$

$$\begin{aligned} \mathcal{A}_X &= [\alpha, \infty) \\ &\quad (\alpha, \infty) \\ &\quad (0, \infty) \end{aligned}$$

$$\|X\|_{\psi_2} \geq 0$$

$$\text{Suppose } X = 0 \text{ w.p. } 1$$

$$\mathbb{E} e^{X^2/K^2} = 1 \quad \forall K > 0$$

$$\mathcal{A}_X = (0, \infty)$$

$$\|X\|_{\psi_2} = 0$$

Suppose $\Pr[X \neq 0] > 0$

$\Rightarrow \exists \epsilon > 0, \delta > 0$

$\Pr[X^2 > \epsilon^2] \geq \delta$

$$\mathbb{E} e^{X^2/k^2} = \int_{-\infty}^{\infty} e^{x^2/k^2} f_X(x) dx$$



$$\geq \int_{x: x^2 \geq \epsilon^2} e^{x^2/k^2} f_X(x) dx$$

$$\geq \int_{x: x^2 \geq \epsilon^2} e^{\epsilon^2/k^2} f_X(x) dx$$

$$= e^{\epsilon^2/k^2} \Pr[X^2 \geq \epsilon^2]$$

$\geq e^{\epsilon^2/k^2} \delta > 2$ for small enough $k > 0$

$\Rightarrow \|X\|_{\psi_2} > 0$

Suppose X & Y are SG (Not necessarily independent)

$$\|X\|_{\psi_2} = K_X$$

$$\|Y\|_{\psi_2} = K_Y$$

$g(x) = e^{x^2}$ is increasing & convex

$$g(\alpha x + (1-\alpha)y) \leq \alpha g(x) + (1-\alpha)g(y)$$
$$0 \leq \alpha \leq 1$$

Want to show:

$$\|X+Y\|_{\psi_2} \leq \|X\|_{\psi_2} + \|Y\|_{\psi_2}$$
$$= K_X + K_Y$$
$$\equiv \mathbb{E} e^{(X+Y)^2 / (K_X + K_Y)^2} \leq 2$$

$$g\left(\frac{|X+Y|}{K_X+K_Y}\right) = e^{(X+Y)^2/(K_X+K_Y)^2}$$

$$\leq g\left(\frac{|X|+|Y|}{K_X+K_Y}\right)$$

$$= g\left(\frac{|X|}{K_X+K_Y} + \frac{|Y|}{K_X+K_Y}\right)$$

$$\leq g\left(\frac{K_X}{K_X+K_Y} \frac{|X|}{K_X} + \frac{K_Y}{K_X+K_Y} \frac{|Y|}{K_Y}\right)$$

convexity

$$\leq \frac{K_X}{K_X+K_Y} g\left(\frac{|X|}{K_X}\right) + \frac{K_Y}{K_X+K_Y} g\left(\frac{|Y|}{K_Y}\right)$$

$$\mathbb{E} \exp\left[\frac{(X+Y)^2}{(K_X+K_Y)^2}\right] \leq \frac{K_X}{K_X+K_Y} \mathbb{E} e^{X^2/K_X^2} + \frac{K_Y}{K_X+K_Y} \mathbb{E} e^{Y^2/K_Y^2} \leq 2$$

$$\Rightarrow \|x+y\|_{\psi_2} \leq k_x + k_y \rightarrow \text{Triangle inequality}$$

$$\Rightarrow \text{If } x_1, x_2, \dots, x_n \text{ are SG with } k_i = \|x_i\|_{\psi_2} \\ \& \alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{R}$$

$$\text{Then } \sum_{i=1}^n \alpha_i x_i \text{ is SG with}$$

$$\left\| \sum_i \alpha_i x_i \right\|_{\psi_2} \leq \sum_{i=1}^n |\alpha_i| k_i$$

Take $X = \alpha$ w.p. 1

$$\mathbb{E} e^{X^2/k^2} = e^{\alpha^2/k^2} = 2 \Rightarrow \frac{\alpha^2}{k^2} = \ln 2$$

$$\|X\|_{\psi_2} = \frac{|\alpha|}{\sqrt{\ln 2}}$$

$X = \alpha$ w.p. 1

$Y = \beta$ w.p. 1

$\alpha > 0$

$\beta > 0$

$$\|X + Y\|_{\psi_2} = \frac{\alpha + \beta}{\sqrt{\ln 2}} = \|X\|_{\psi_2} + \|Y\|_{\psi_2}$$

Take X is SG & $Y = X$ w.p. 1

$$\begin{aligned} \|X + Y\|_{\psi_2} &= \|2X\|_{\psi_2} = 2\|X\|_{\psi_2} \\ &= \|X\|_{\psi_2} + \|Y\|_{\psi_2} \end{aligned}$$

What if X & Y are independent? & $EX = 0$
 $EY = 0$

$$E e^{tX} \leq \exp \left[c t^2 \|X\|_{\Psi_1}^2 \right]$$

$$E e^{tY} \leq \exp \left[c t^2 \|Y\|_{\Psi_2}^2 \right]$$

$$E e^{t(X+Y)} \leq E e^{tX} E e^{tY}$$

$$\leq \exp \left[c t^2 (\|X\|_{\Psi_1}^2 + \|Y\|_{\Psi_2}^2) \right]$$

$$\|X+Y\|_{\Psi}^2 \leq \|X\|_{\Psi_1}^2 + \|Y\|_{\Psi_2}^2$$

Hoeffding's inequality : If $X_1, \dots, X_n$ are independent
& $\|X_i\|_{\psi_2} \leq K_i$, $\alpha_1, \dots, \alpha_n \in \mathbb{R}$
 $\mathbb{E} X_i = 0$

Then,

$$P_n \left[\left| \sum_{i=1}^n \alpha_i X_i \right| \geq \delta \right] \leq 2 \exp \left[- \frac{c \delta^2}{\sum_{i=1}^n \alpha_i^2 K_i^2} \right]$$

If $X \in [-\alpha, \alpha]$ w.p. 1 for some $\alpha > 0$

$$\Pr[|X| > \alpha] = 0$$

$$\mathbb{E}|X|^p \leq \alpha^p \quad \forall p \geq 1$$
$$\leq \alpha^p p^{p/2}$$

$$\left(\mathbb{E}|X|^p\right)^{1/p} \leq \alpha \sqrt{p}$$

Thm
 $\Rightarrow$

$$\|X\|_{\psi_2}$$

$$\leq c \alpha$$

for some
universal
constant

Application

Hypothesis testing

$$X \in \{0, 1\}$$

the hypothesis

$$X = \text{Ber}(\frac{1}{2})$$

fair
coin

biased

coin

$$\text{If } X = 0, \\ 1,$$

$$Y_1, \dots, Y_n \sim \text{iid } p_0$$

$$Y_1, \dots, Y_n \sim \text{iid } p_1$$

Assume: We know p_0 & p_1 .

$$p(y) \geq p_{\min} > 0$$

$$p_1(y) \geq p_{\min} > 0$$

Given $Y_1, \dots, Y_n$ decide if $X = 0$ or 1

$$P_n[\hat{X}(Y_1, \dots, Y_n) \neq X]$$

$$\hat{X} = \begin{cases} 1 & \text{if } P_n[Y^n|1] \geq P_n[Y^n|0] \\ 0 & \text{else} \end{cases}$$

$$= \begin{cases} 1 & \text{if } \log \frac{P_n[Y^n|1]}{P_n[Y^n|0]} \geq 0 \\ 0 & \text{else} \end{cases}$$

$$= \begin{cases} 1 & \text{if } \log \frac{\prod_{i=1}^n p_1(Y_i)}{\prod_{i=1}^n p_0(Y_i)} \geq 0 \\ 0 & \text{else} \end{cases}$$

$$\log \left[\prod_{i=1}^n \frac{p_1(y_i)}{p_0(y_i)} \right] = \sum_{i=1}^n \log \left(\frac{p_1(y_i)}{p_0(y_i)} \right)$$

$$\log p_{\min} \leq \log \left(\frac{p_1(y)}{p_0(y)} \right) \leq \log \frac{1}{p_{\min}}$$

$$P_n[\hat{X} \neq X] = \frac{1}{2} P_n[\hat{X} \neq X | X=0] + \frac{1}{2} P_n[\hat{X} \neq X | X=1]$$

Analysis

$$P_n[\hat{X} \neq x \mid x=1]$$

$$= P_n[\hat{X} = 0 \mid x=1]$$

$$= P_n \left[\log \frac{\prod_{i=1}^n p_1(y_i)}{\prod_{i=1}^n p_0(y_i)} < 0 \mid x=1 \right]$$

$y_1, \dots, y_n \sim \text{iid}(p_1)$

$$Z_n = \sum_{i=1}^n \log \frac{p_1(y_i)}{p_0(y_i)}$$

$$\begin{aligned} \mathbb{E}_{p_1} Z_n &= \sum_{i=1}^n \mathbb{E}_{p_1} \left[\log \frac{p_1(y_i)}{p_0(y_i)} \right] = \sum_{i=1}^n \sum_{y \in \mathcal{Y}} p_1(y) \log \frac{p_1(y)}{p_0(y)} \\ &= \sum_{i=1}^n D(p_1 \| p_0) \end{aligned}$$

$$= n D(p, \| p_0)$$

$$P_n[\hat{X} \neq X | X = 1] = P_n[Z_n < 0 | X = 1]$$

$$= P_n[Z_n - \mathbb{E} Z_n < -\mathbb{E} Z_n | X = 1]$$

$$= P_n[Z_n - \mathbb{E} Z_n < -\delta | X = 1]$$

$$\text{Hoeffding} \leq 2 \exp \left[- \frac{c (\mathbb{E} Z_n)^2}{\sum_{i=1}^n K_i^2} \right] \quad \left| \log \frac{p_i(y)}{p_0(y)} \right| \leq \log \frac{1}{p_{\min}}$$

$$= 2 \exp \left[- c \frac{n^2 (D(p, \| p_0))^2}{n \left(\log \frac{1}{p_{\min}} \right)^2} \right]$$

$$= 2 \exp \left[- \frac{n c (\mathcal{D}(p_1 \| p_0))^2}{\left(\log \frac{1}{p_{mn}} \right)^2} \right]$$

THEOREM Let X be a random variable

The following are equivalent (here, $\frac{K_i}{K_j}$ can be chosen absolute constant that does not depend on f_X)

$$\textcircled{1} \quad \Pr[|X| \geq \delta] \leq 2 \exp\left[-\frac{\delta^2}{K_1^2}\right] \quad \forall \delta \geq 0$$

$$\textcircled{2} \quad \left(\mathbb{E}|X|^p\right)^{1/p} \leq K_2 \sqrt{p} \quad p \geq 1$$

$$\textcircled{3} \quad \mathbb{E} e^{t^2 X^2} \leq e^{K_3^2 t^2} \quad \forall t: |t| \leq \frac{1}{K_3}$$

$$\textcircled{4} \quad \mathbb{E} e^{X^2/K_4^2} \leq 2$$

$$\text{If } \mathbb{E}X = 0,$$

$$\textcircled{5} \quad \mathbb{E} e^{tX} \leq e^{K_5^2 t^2} \quad \forall t \in \mathbb{R}$$

$$\|X\|_{\psi_2} = \inf \{K > 0: \mathbb{E} e^{X^2/K^2} \leq 2\}$$

Exponential & Poisson

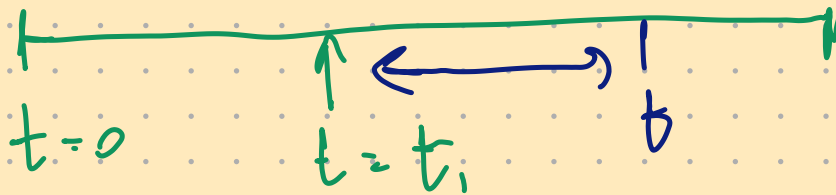
Waiting/ Service times :

- ① Arrival of bus
- ② Time for completion of job

Assumption : Memoryless property

$$G(t) = \Pr[T > t] = 1 - F_T(t)$$

$$G(t_1 + t_2) = G(t_1) \times G(t_2)$$



T : rv that corresponds to the waiting time

Assume $f_T(t) = \lambda e^{-\lambda t}$

$$G(t) = Pn[T > t] = \int_t^{\infty} \lambda e^{-\lambda z} dz = e^{-\lambda t}$$

$$G(t_1 + t_2) = e^{-\lambda(t_1 + t_2)} = G(t_1) G(t_2)$$

"

$$Pn[T > t_1 + t_2]$$

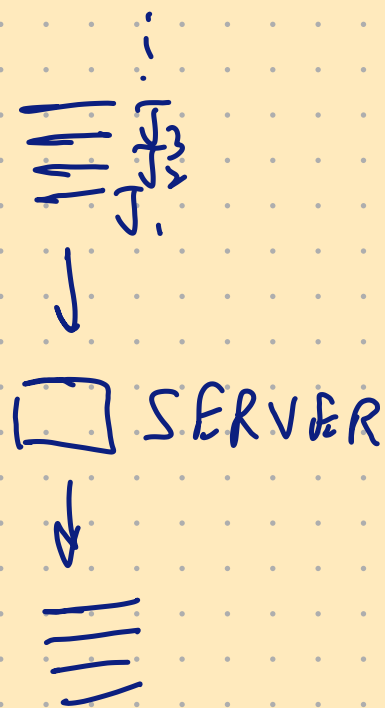
$$\frac{Pn[T > t_1]}{Pn[T > t_2]}$$

"

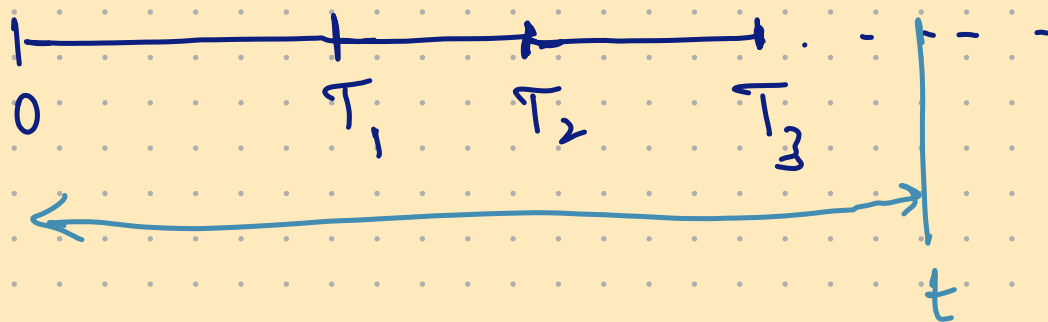
$$Pn[T > t_1]$$

$$Pn[T > t_1 + t_2 | T > t_1]$$

FACT: The exponential distribution is the only distribution with the memoryless property



Assume that server takes $T \sim \text{EXP}(\lambda)$
 & jobs are independent



N_t



of jobs completed in
 time t

Claim: $N_t \sim \text{Poisson}(\lambda t)$

$$P[N_t = n] = \frac{e^{-\lambda t} (\lambda t)^n}{n!}$$

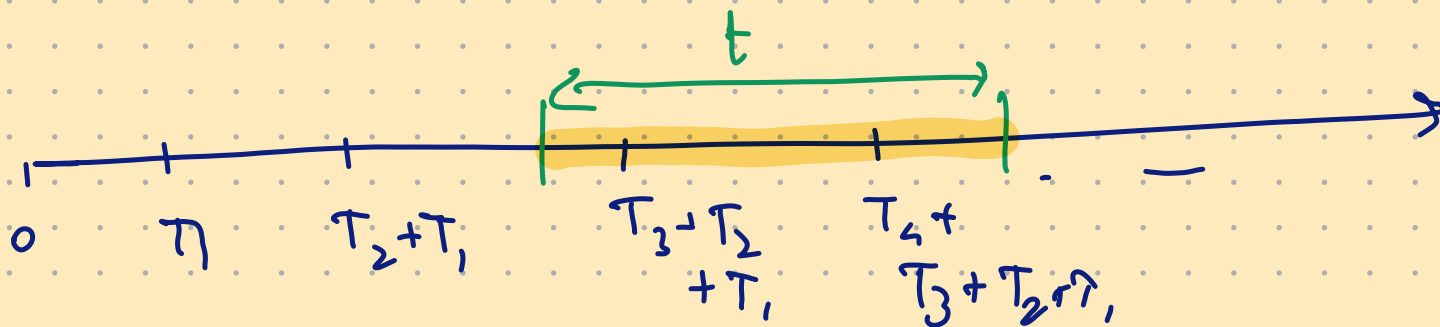
$$\begin{aligned}
 G(t) &= \Pr[\tau_1 > t] = \Pr[N_t = 0] \\
 &= e^{-\lambda t} \frac{(\lambda t)^0}{0!} \\
 &= e^{-\lambda t}
 \end{aligned}$$

$$\mathbb{E} N_t = \lambda t$$

$$\Pr[\tau_i > t] = e^{-\lambda t}$$

"

$$\Pr[N_{t+z} = 0 \mid N_t = 0] = e^{-\lambda z}$$



Poisson process : For any interval $[\alpha, \beta]$,
 $N_{[\alpha, \beta]}$ = # of jobs completed in $[\alpha, \beta]$
 $\sim \text{Poisson}(\lambda(\beta - \alpha))$

$N_{[\alpha, \beta]}$ is independent of $N_{[\alpha', \beta']}$
if $[\alpha', \beta'] \cap [\alpha, \beta] = \emptyset$

$$\mathbb{E} N_t = \lambda t$$

$$P_n(|N_t - \lambda t| > \sigma) = ?$$

$$X \sim \text{Exp}(\lambda)$$

$$\text{Pr}[X > t] = e^{-\lambda t}$$

$$\log \text{Pr}[X > t] = -\lambda t$$

$$X \sim \text{Poisson}(\alpha)$$

$$\alpha > 0$$

$$p_X(i) = \frac{\alpha^i}{i!} e^{-\alpha}$$

$$\mathbb{E} e^{tX} = \sum_{i=0}^{\infty} \frac{\alpha^i}{i!} e^{-\alpha} e^{ti} = \sum_{i=0}^{\infty} \frac{(\alpha e^t)^i}{i!} e^{-\alpha}$$

$$= \exp[\alpha e^t - \alpha]$$

$$\begin{aligned}
 P_n[X \geq \alpha + \delta] &\leq \frac{\exp[\alpha e^t - \alpha]}{\exp[t(\alpha + \delta)]} \\
 &= \exp[\alpha e^t - \alpha - t(\alpha + \delta)] \\
 &\quad \forall t > 0
 \end{aligned}$$

$$g(t) = \alpha e^t - \alpha - t(\alpha + \delta)$$

$$t^* = \ln\left(\frac{\alpha + \delta}{\alpha}\right)$$

$$\begin{aligned}
 P_n[X \geq \alpha + \delta] &\leq \exp\left[\cancel{\alpha} \frac{(\cancel{\alpha} + \delta)}{\cancel{\alpha}} - \cancel{\alpha} - (\alpha + \delta) \ln\left(\frac{\alpha + \delta}{\alpha}\right)\right] \\
 &= \exp\left[\delta - (\alpha + \delta) \ln\left(\frac{\alpha + \delta}{\alpha}\right)\right]
 \end{aligned}$$

$$= \exp \left(\alpha \frac{\delta}{\alpha} - \alpha \left(1 + \frac{\delta}{\alpha} \right) \ln \left(1 + \frac{\delta}{\alpha} \right) \right)$$

$$= \exp \left[-\alpha \underbrace{\left(\left(1 + \frac{\delta}{\alpha} \right) \ln \left(1 + \frac{\delta}{\alpha} \right) - \frac{\delta}{\alpha} \right)} \right]$$

$$g(u) = (1+u) \ln(1+u) - u$$

$$g(u) \geq \begin{cases} \frac{u^2}{4} & \text{if } 0 \leq u \leq 1 \\ (2 \ln 2 - 1) u & , \quad u > 1 \end{cases}$$

$$\textcircled{1} \quad \underbrace{g(u) - \frac{u^2}{4}}_{h(u)} \geq 0 \quad \text{for} \quad 0 \leq u \leq 1$$

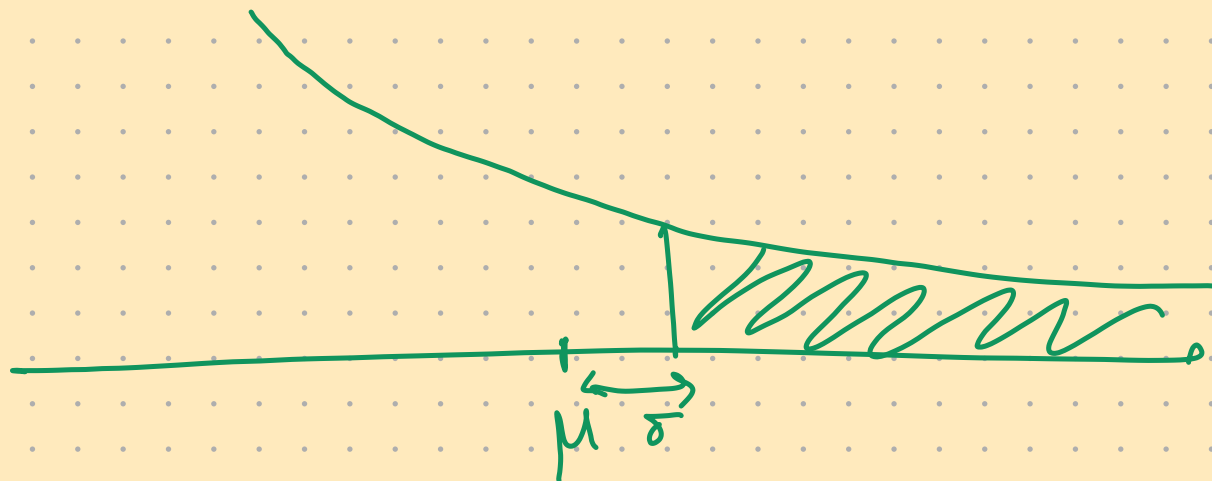
$$h(0) = g(0) - \frac{0}{4} = 0$$

$$h'(u) = \ln(1+u) - \frac{u}{2} \geq 0 \quad u \in [0, 1]$$

$$P_n[X \geq \alpha + \delta] \leq \begin{cases} \exp\left[-\frac{\delta^2}{4\alpha}\right] & 0 \leq \frac{\delta}{\alpha} \leq 1 \\ \exp[-c\delta] & \delta > \alpha \end{cases}$$

Small deviations
are subgaussian

Large deviations
are exponential



$$P_n[X \geq \mu + \sigma]$$

$$P_n[|X - \mu| \geq \sigma] \leq$$

Theorem

Let X be a n.v. The following are equivalent (Here, we can choose K_i/k_j to be universal constants)

$$\textcircled{1} \quad P_n[|X| \geq t] \leq 2 \exp\left[-\frac{t}{K_1}\right] \quad \forall t \geq 0$$

$$\textcircled{2} \quad (\mathbb{E}|X|^p)^{1/p} \leq K_2 p \quad \forall p \geq 1$$

$$\textcircled{3} \quad \mathbb{E} e^{t|X|} \leq e^{K_3 t} \quad \forall t \geq 0 \quad 0 \leq t \leq \frac{1}{K_3}$$

$$\textcircled{4} \quad \mathbb{E} e^{|X|/K_4} \leq 2$$

Additionally, if $\mathbb{E} X = 0$,

$$\textcircled{5} \quad \mathbb{E} e^{tX} \leq e^{K_5^2 t^2}, \quad |t| \leq \frac{1}{K_5}$$

Subexponential norm:

$$\|X\|_{\psi_1} = \inf \{ K > 0 : \mathbb{E} e^{|X|/K} \leq 2 \}$$

$$\|X\|_{\psi_2} = \inf \{ K > 0 : \mathbb{E} e^{X^2/K^2} \leq 2 \}$$

* $\text{Exp}(\lambda)$ is subexponential but not subgaussian

* From prop 2, every subgaussian RV is subexponential

Suppose that X is SG

$$\mathbb{E} e^{X^2/K^2} \leq 2$$

$$Y = X^2 \quad \mathbb{E} e^{Y/K^2} \leq 2$$

$$\|X^2\|_{\psi_1} = \|X\|_{\psi_2}^2$$

* If X is SG, then X^2 is SE.

Ex: If X & Y are Subgaussian (not necessarily independent),
then XY is subexponential

$$\|XY\|_{\psi_1} \leq \|X\|_{\psi_2} \|Y\|_{\psi_2}$$

* If X & Y are SE

$$\|\alpha X + \beta Y\|_{\psi_1} \leq |\alpha| \|X\|_{\psi_1} + |\beta| \|Y\|_{\psi_1}$$

* Ex: X is SG $\mathbb{E}X$

$$\|X - \mathbb{E}X\|_{\psi_2} \leq C \|X\|_{\psi_2}$$

X is SE

$$\|X - \mathbb{E}X\|_{\psi_1} \leq C \|X\|_{\psi_1}$$

* If X is SG & $\mathbb{E}X = 0$, then

$$\|X\|_{\psi_2}^2 \geq C \text{Var}(X)$$

Bernstein's inequality: $X_1, \dots, X_n$ are independent

s.t. $\mathbb{E} X_i = 0 \ \forall i$, & $\underline{a} \in \mathbb{R}^n$ (is constant)

Then, $\forall \delta > 0$

$$P_n \left[\left| \sum_{i=1}^n a_i X_i \right| \geq \delta \right] \leq 2 \exp \left[-c \min \left\{ \frac{\delta^2}{\sum_{i=1}^n a_i^2 \|X_i\|_{\Psi_1}^2}, \frac{\delta}{\max_i a_i \|X_i\|_{\Psi_1}} \right\} \right]$$

If $\underline{X} = (X_1, \dots, X_n)$ has independent SG components,

then $\|\underline{X}\|_2^2$ is SE $\Rightarrow \|\underline{X}\|_2^2 - \mathbb{E} \|\underline{X}\|_2^2$ is SE

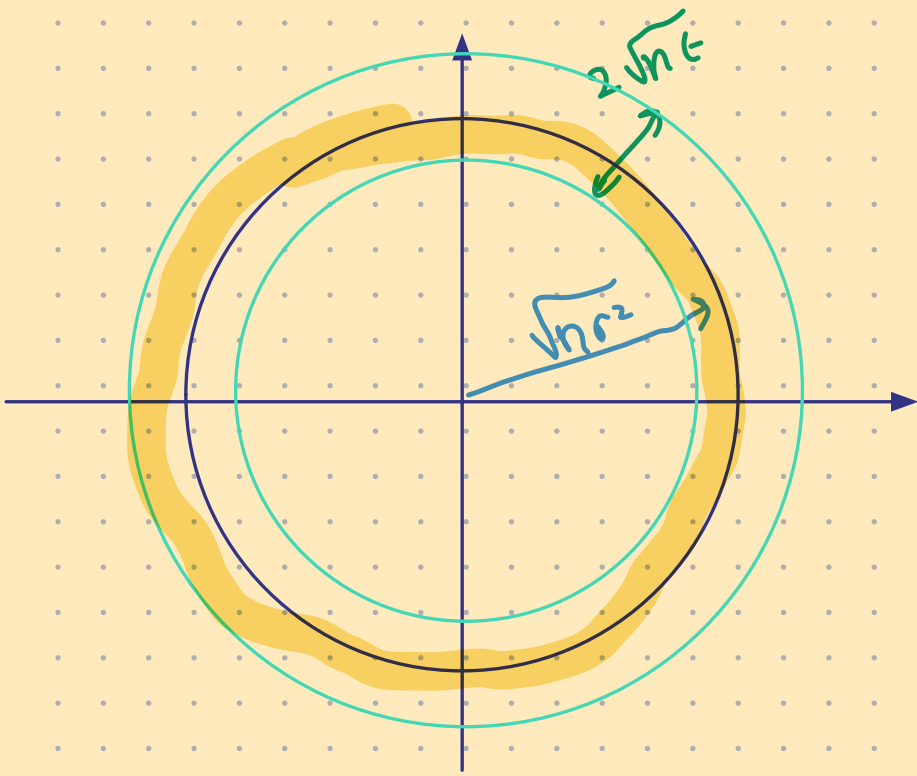
$$\Pr \left[\left| \|\underline{X}\|_2^2 - \mathbb{E} \|\underline{X}\|_2^2 \right| \geq \delta \right] \leq 2 \exp \left[-c \min \left\{ \frac{\delta^2}{\sum_i \|X_i\|_{\Psi_2}^4}, \frac{\delta}{\max_i \|X_i\|_{\Psi_2}^2} \right\} \right]$$

Assume $\underline{X}$ is an iid $\mathcal{N}(0, \sigma^2)$

$$\mathbb{E} \|\underline{X}\|_2^2 = \mathbb{E} \sum_i X_i^2 = n \sigma^2$$

$$\|X_i\|_{\Psi_2}^2 = c' \sigma^2$$

$$\Pr \left[\left| \|\underline{X}\|_2^2 - n \sigma^2 \right| \geq n \epsilon \right] \leq 2 \exp \left[-c \min \left\{ \frac{n^2 \epsilon^2}{c' n \sigma^2}, \frac{n \epsilon}{\sigma^2} \right\} \right]$$



$$P_n \left[\underline{X} \notin S_{2\sqrt{n\epsilon}}(\sqrt{n\sigma^2}) \right]$$

$$\Rightarrow P_n \left[\|\underline{X}\|_2^2 < n\sigma^2(1-\epsilon) \right] \rightarrow 0 \text{ as } n \rightarrow \infty$$

If $\underline{X}$ has iid SG components, $\mathbb{E}X_i = 0$
 $\mathbb{E}X_i^2 = 1$

$$K_i = \|\underline{X}_i\|_{\psi_2}$$

$Y = \|\underline{X}\|_2 - \sqrt{n}$ is SG

$$\|Y\|_{\psi_2} \leq c \max_i K_i^2 = c \max_i \|\underline{X}_i\|_{\psi_2}^2$$

Proof: $X_i^2 - 1$ is SG

$$\|X_i^2 - 1\|_{\psi_1} \leq c \|\underline{X}_i\|_{\psi_2}^2$$

use Bernstein's inequality

$$\begin{aligned} P_n \left[\left| \|X\|_2^2 - n \right| \geq n\delta \right] &\leq 2 \exp \left[-c \min \left[\frac{n^2 \delta^2}{\sum_i K_i^4}, \frac{n\delta}{\max_i K_i^2} \right] \right] \\ &\leq 2 \exp \left[-c \times \min \left[\frac{n^2 \delta^2}{n K_{\max}^4}, \frac{n\delta}{K_{\max}^2} \right] \right] \end{aligned}$$

$$\left(\begin{array}{l} K_{\max} \geq c' \text{ for some universal const } c' \\ K_{\max}^4 \geq c'' K_{\max}^2 \end{array} \right)$$

$$\leq 2 \exp \left[-\frac{c'' n}{K_{\max}^4} \min \{ \delta^2, \delta \} \right]$$

— ①

Observation : for any $z \in \mathbb{R}$.

Ex : $\underbrace{|z-1| \geq \delta}_A \Rightarrow \underbrace{|z^2-1| \geq \max\{\delta, \delta^2\}}_B$

$$P(A) \leq P(B)$$

$$P_n \left[|\|X\|_2 - \sqrt{n}| \geq \sqrt{n} \epsilon \right] \leq P_n \left[|\|X\|_2^2 - n| \geq n \max\{\epsilon, \epsilon^2\} \right]$$

Using ①,

$$\leq 2 \exp \left[- \frac{cn}{K_{\max}^4} \min\{\delta, \delta^2\} \right]$$

$$\delta = \max\{\epsilon, \epsilon^2\}$$

$$\min\left\{\max\{\epsilon, \epsilon^2\}, \max\{\epsilon^2, \epsilon^q\}\right\} = \epsilon^2$$

$$\forall \epsilon' > 0, \\ P_n\left[|\|X\|_2 - \sqrt{n}| \geq \sqrt{n} \epsilon\right] \leq 2 \exp\left[-\frac{c n \epsilon^2}{K_{\max}^4}\right]$$

$$P_n\left[|\|X\|_2 - \sqrt{n}| \geq \delta'\right] \leq 2 \exp\left[-\frac{c (\delta')^2}{K_{\max}^4}\right]$$

Using prop 1 of SG n.v, ($X_1, \dots, X_n$ are indep SG)

$$\|\|X\|_2 - \sqrt{n}\|_{\psi_2} \leq c K_{\max}^2$$

APPLICATION: Dimensionality reduction

$$(\underline{a}_1, \underline{a}_2, \dots, \underline{a}_N) \quad \underline{a}_i \in \mathbb{R}^D \quad D \gg N$$

Complexity of computing p.w distances = $O(N^2 D)$

① $g: \mathbb{R}^D \rightarrow \mathbb{R}^d \quad d \ll D$

② compute distances in $\mathbb{R}^d$
 $g(\underline{a}_i) \quad g(\underline{a}_j)$

Want g s.t

$$(1-\epsilon) \|\underline{a}_i - \underline{a}_j\|_2 \leq \|g(\underline{a}_i) - g(\underline{a}_j)\|_2 \leq (1+\epsilon) \|\underline{a}_i - \underline{a}_j\|_2$$

(ϵ - isometry)

(Johnson - Lindenstrauss lemma) if $g(\underline{a}) = \sum \underline{a}$

$\underline{X}$ iid SG

$$\& d = \Omega\left(\frac{K^4}{\epsilon^2} \log \frac{N}{\sqrt{\delta}}\right)$$

$$\Pr[\underline{X} \text{ is an } \epsilon\text{-isometry}] \geq 1 - \delta$$

Lemma :

If $X_1, \dots, X_n$ are independent SG with $\mathbb{E}X_i = 0$

$\text{Var}(X_i) = 1$, $\|X_i\|_{\psi_2} = K_i$, then

$$\|\underline{X}\|_2 - \sqrt{n} \leq c K_{\max}^2$$

$$\downarrow$$
$$\max_i K_i^2$$

Hoeffding inequality:

$$P_n[|\|\underline{x}\|_2 - \sqrt{n}| \geq \sqrt{n}\epsilon] \leq 2 \exp\left[-\frac{n\epsilon^2 c}{k_{\max}^4}\right]$$

— ①

$$\underline{u}_i = g(\underline{a}_i) = \underline{\sum} \underline{a}_i$$

$$\underline{u}_i = \begin{bmatrix} \sum_{l=1}^D X_{1l} a_{il} \\ \sum_{l=1}^D X_{2l} a_{il} \\ \vdots \\ \sum_{l=1}^D X_{dl} a_{il} \end{bmatrix}$$

$$\begin{aligned} \|\underline{u}_i - \underline{u}_j\|_2 &= \|g(\underline{a}_i) - g(\underline{a}_j)\|_2 \\ &\approx \|\underline{a}_i - \underline{a}_j\| \end{aligned}$$

$$\underline{u}_i - \underline{u}_j = \underline{X} (\underline{a}_i - \underline{a}_j) = \begin{bmatrix} \sum_{l=1}^D X_{1l} (a_{il} - a_{jl}) \\ \vdots \\ \sum_{l=1}^D X_{dl} (a_{il} - a_{jl}) \end{bmatrix}$$

$$\mathbb{E} \|\underline{u}_i - \underline{u}_j\|^2 = \mathbb{E} \sum_{m=1}^d \left(\sum_{l=1}^D X_{ml} (a_{il} - a_{jl}) \right)^2$$

$$X_{ml} \sim \text{iid } \mathcal{GN} \quad \mathbb{E} X_{ml} = 0$$

$$\text{Var}(X_{ml}) = \sigma^2$$

$$\mathbb{E} \|\underline{u}_i - \underline{u}_j\|^2 = \sum_{m=1}^d \sum_{l=1}^D \sigma^2 (a_{il} - a_{jl})^2$$

$$= \sum_{m=1}^d \sigma^2 \| \underline{a}_i - \underline{a}_j \|^2$$

$$= d \sigma^2 \| \underline{a}_i - \underline{a}_j \|^2$$

Choose $\sigma^2 = \frac{1}{d}$, $\sigma^2 = \frac{1}{\sqrt{d}}$

$$\underline{X} \sim \text{iid } SG \quad \|X_{m1}\|_{\Psi_2} = K$$

$$E X_{m1} = 0$$

$$\text{Var}(X_{m1}) = \frac{1}{d}$$

$$\underline{X} = \frac{1}{\sqrt{d}} \underline{Y}$$

$$Y_{m1} \sim SG, \text{ iid}$$

$$\|Y_{m1}\|_{\Psi_2} = K', \quad E Y_{m1} = 0$$

$$\text{Var}(Y_{m1}) = 1$$

$$E \left\| \frac{1}{\sqrt{d}} \sum \underline{a}_i - \frac{1}{\sqrt{d}} \sum \underline{a}_j \right\|_2^2 = \|\underline{a}_i - \underline{a}_j\|^2$$

$$P_n \left[\left| \|g(\underline{a}_i) - g(\underline{a}_j)\|_2 - \|\underline{a}_i - \underline{a}_j\|_2 \right| \geq \epsilon \|\underline{a}_i - \underline{a}_j\| \right]$$

for any $i \neq j$

fix $i \neq j$

$$\begin{aligned} & P_n \left[\left| \|g(\underline{a}_i) - g(\underline{a}_j)\| - \|\underline{a}_i - \underline{a}_j\| \right| \geq \epsilon \|\underline{a}_i - \underline{a}_j\| \right] \\ &= P_n \left[\left| \left\| \frac{1}{\sqrt{d}} \sum (\underline{a}_i - \underline{a}_j) \right\|_2 - \|\underline{a}_i - \underline{a}_j\|_2 \right| \geq \epsilon \|\underline{a}_i - \underline{a}_j\| \right] \\ &> P_n \left[\left| \left\| \underbrace{\frac{1}{\|\underline{a}_i - \underline{a}_j\|}}_{\substack{\in \mathbb{R}^d \\ \text{iid } 0 \text{ mean, Var} = 1, \text{ SG } K}} \sum (\underline{a}_i - \underline{a}_j) \right\|_2 - \sqrt{d} \right| \geq \epsilon \sqrt{d} \right] \end{aligned}$$

$$\leq 2 \exp \left[-c \frac{d \epsilon^2}{K^4} \right]$$

$$P_n \left[\left| \|g(a_i) - g(a_j)\|_2 - \|a_i - a_j\|_2 \right| \geq \epsilon \|a_i - a_j\| \text{ for some } i \neq j \right]$$

$$P_n \left[g \text{ is not an } \epsilon\text{-isometry for } \{a_1, \dots, a_n\} \right]$$

$$\leq \binom{N}{2} 2 \exp \left[-c \frac{d \epsilon^2}{K^4} \right]$$

$$\leq 2N^2 \exp \left[-c \frac{d \epsilon^2}{K^4} \right] \leq \delta$$

$$\frac{2N^2}{\delta} \leq \exp\left[\frac{cd\epsilon^2}{K^4}\right]$$

$$d \geq c' \frac{K^4}{\epsilon^2} \ln\left(\frac{2N^2}{\delta}\right)$$

$$= c'' \frac{K^4}{\epsilon^2} \ln\left(\frac{N}{\sqrt{\delta}}\right)$$

BOUNDING THE VARIANCE

Let $Z = \sum_{i=1}^n X_i$, $X_1, \dots, X_n$ independent

$$\text{Var}(Z) = \sum_{i=1}^n \text{Var}(X_i) = O(n)$$

Want to bound the variance for a (more) general

$$Z = g(X_1, \dots, X_n), \quad X_1, \dots, X_n \text{ independent}$$

$$\mathbb{E}[Z | X_1, \dots, X_i] = \mathbb{E}[g(X^n) | X_1, \dots, X_i]$$

$$= \int_{\mathbb{R}^{n-i}} g(X_1, \dots, X_i, x_{i+1}, \dots, x_n) f_{X_{i+1}, \dots, X_n | X_1, \dots, X_i}(x_{i+1}, \dots, x_n) dx_{i+1} \dots dx_n$$

Define $\Delta_i = E[Z | X_1, \dots, X_i] - E[Z | X_1, \dots, X_{i-1}]$

$$\sum_{i=1}^n \Delta_i = E[Z | X_1, \dots, X_n] - E[Z]$$

$$= Z - EZ$$

$$\text{Var}(Z) = E\left[\left(\sum_{i=1}^n \Delta_i\right)^2\right]$$

$$= E\left[\sum_{i=1}^n \Delta_i^2\right] + E\left[\sum_i \sum_{j \neq i} \Delta_i \Delta_j\right]$$

$$= E\left[\sum_{i=1}^n \Delta_i^2\right] + 2 E\left[\sum_i \sum_{j > i} \Delta_i \Delta_j\right]$$

fix $i, j > i$

$$E \Delta_i \Delta_j$$

$$\mathbb{E} \left[\underbrace{\left(\mathbb{E}[Z | X_1, \dots, X_i] - \mathbb{E}[Z | X_1, \dots, X_{i-1}] \right)}_{\text{fn of } X_1, \dots, X_i} \underbrace{\left(\mathbb{E}[Z | X_1, \dots, X_j] - \mathbb{E}[Z | X_1, \dots, X_{j-1}] \right)}_{X_1, \dots, X_j} \right]$$

$$\mathbb{E}[\Delta_i \Delta_j] = \mathbb{E} \left[\underbrace{\mathbb{E}[\Delta_i \Delta_j | X_1, \dots, X_i]}_{\substack{\text{wrt} \\ f_{X_1, \dots, X_i}}} \right]$$

$\downarrow$ $\downarrow$
 wrt $f_{X_{i+1}, \dots, X_j | X_1, \dots, X_i}$

$$= \mathbb{E} \left[\Delta_i \underbrace{\mathbb{E}[\Delta_j | X_1, \dots, X_i]}_0 \right]$$

$$= 0$$

$$\mathbb{E}[\Delta_j | X_1, \dots, X_i] = \mathbb{E} \left[\mathbb{E}(Z | X_1, \dots, X_j) - \mathbb{E}(Z | X_1, \dots, X_{j-1}) \mid X_1, \dots, X_i \right]$$

$$= \mathbb{E} \left[\mathbb{E}(Z | X_1, \dots, X_j) \mid X_1, \dots, X_i \right] - \mathbb{E} \left[\mathbb{E}(Z | X_1, \dots, X_{j-1}) \mid X_1, \dots, X_i \right]$$

$$= \mathbb{E}[Z | X_1, \dots, X_i] - \mathbb{E}[Z | X_1, \dots, X_i] = 0$$

$\downarrow$
 $f_{X_{i+1}, \dots, X_n | X_1, \dots, X_i}$

$$\mathbb{E}[W | X_1, \dots, X_i] = \int f_{X_{i+1}, \dots, X_n | X_1, \dots, X_i}(x^n) \underbrace{W}_{\mathbb{E}(Z | X_1, \dots, X_j)} dx_{i+1} \dots dx_n$$

$$= \int f_{X_{i+1} \cdots X_n | X_1 \cdots X_i} (x_{i+1} - x_n) \int f_{X_{j+1} \cdots X_n | X_i \cdots X_j} g(x_i - x_n) dx$$

$$Z = g(X_1 \cdots X_n)$$

$$\text{Var}(Z) = \sum_{i=1}^n \mathbb{E} \Delta_i^2$$

$$\text{where } \Delta_i = \mathbb{E}[Z | X_1 \cdots X_i] - \mathbb{E}[Z | X_1 \cdots X_{i-1}]$$

Assume that $X_1, \dots, X_n$ are independent

$$\Delta_i^2 = \left(E(Z | X_1, \dots, X_i) - E(Z | X_1, \dots, X_{i-1}) \right)^2$$

$$= \left\{ E \left(Z - \underbrace{E^{(i)} Z}_{\text{green}} \mid X_1, \dots, X_i \right) \right\}^2$$

$$\int f_{X_i}(x_i) g(X_1, \dots, X_{i-1}, x_i, X_{i+1}, \dots, X_n) dx_i$$

$$\leq E \left[\left(Z - E^{(i)}(Z) \right)^2 \right]$$

Efron-Stein Inequality

For any $Z = g(X_1, \dots, X_n)$ where $X_1, \dots, X_n$ are independent,

$$\text{Var}(Z) \leq \sum_{i=1}^n \mathbb{E} \left[(Z - \mathbb{E}^{(i)} Z)^2 \right] =: V$$

Define $Z'_i = g(X_1, \dots, X_{i-1}, X'_i, X_{i+1}, \dots, X_n)$

$\downarrow$
 X_i replaced with iid copy

$$V = \frac{1}{2} \sum_{i=1}^n \mathbb{E} \left[(Z - Z'_i)^2 \right]$$

$$= \sum_{i=1}^n \mathbb{E} \left[(Z - Z'_i)_+^2 \right] = \sum_{i=1}^n \mathbb{E} \left[(Z - Z'_i)_-^2 \right]$$

where

$$\kappa_+ = \max\{\kappa, 0\}$$

$$\kappa_- = \max\{-\kappa, 0\}$$

$$V = \inf_{z_i} \sum_{i=1}^n \mathbb{E} (z - z_i)^2$$



κ_V that depends

on $X_i, X_{i-1}, X_{i+1}, \dots, X_n$

Functions with bounded differences

$g: \mathcal{X}^n \rightarrow \mathbb{R}$ has the BDP if
 $\exists c_1, \dots, c_n$ s.t.

$$\max_{x_1, \dots, x_n, x'_i} |g(x_1, \dots, x_n) - g(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n)| \leq c_i$$



replace x_i with x'_i

Eq: $g(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n x_i, \quad x_i \in [-1, 1]$

$$|g(x_1, \dots, x_n) - g(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n)| = \left| \frac{x_i - x'_i}{n} \right| \leq \frac{2}{n}$$

for any g with BDP $(c_1, \dots, c_n)$:

$$\text{Var}(z) \leq \inf_{z_i} \sum_{i=1}^n \mathbb{E}(z - \underline{z}_i)^2$$



dep only on $X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n$

$$z_i = g(X_1, \dots, X_{i-1}, x_i, X_{i+1}, \dots, X_n)$$

$$\Rightarrow \inf_{z_i} \mathbb{E}(z - z_i)^2 \leq c_i^2$$

$$\text{Var}(z) \leq \sum_{i=1}^n c_i^2$$

* Can show that Y_g has BDP

$$\text{Var}(Z) \leq \frac{1}{4} \sum_{i=1}^n c_i^2$$

* Can use Chebyshev ineq:

$$\Pr[|Z - \mathbb{E}Z| \geq \sigma] \leq \frac{\frac{1}{4} \sum_{i=1}^n c_i^2}{\sigma^2}$$

* McDiarmid's inequality

$$\Pr[|Z - \mathbb{E}Z| \geq \sigma] \leq 2 \exp\left[-\frac{2\sigma^2}{\sum_{i=1}^n c_i^2}\right]$$

Example : $x_1, \dots, x_n \in [0, 1]$

↓
time req to complete i th task
(resources)

Assume each server has capacity 1

$$x_1 = \frac{1}{3}, x_2 = \frac{1}{2}, x_3 = \frac{9}{10}, x_4 = \frac{2}{3}, x_5 = \frac{3}{7}$$

Let $g(x_1, \dots, x_n)$ be the min # of servers req.
to complete $x_1, \dots, x_n$

$x_1, \dots, x_n$ are iid

$$\text{Var}(g(x_1, \dots, x_n)) \leq ?$$

Given $x_1, \dots, x_n$, if we replace x_i with x'_i

$$|g(x_1, \dots, x_n) - g(x_1, \dots, x'_i, x_{i+1}, \dots, x_n)| \leq 1$$

$$\text{Var}(g(x_1, \dots, x_n)) \leq \frac{1}{4} \sum_{i=1}^n 1^2 = \frac{n}{4}$$

$$\text{If } \mathbb{E}Z \propto n$$

$$P_n[|Z - \mathbb{E}Z| \geq \delta \mathbb{E}Z] \leq \frac{n}{4 \delta^2 (\mathbb{E}Z)^2} = O\left(\frac{1}{\delta^2 n}\right)$$

Largest eigenvalue of a symmetric random matrix

$\underline{X} : n \times n$ random matrix

$$X_{ij} = X_{ji} \quad \forall i, j$$

$(X_{ij} : 1 \leq i \leq j \leq n)$ are iid

$$X_{ij} \in [-1, 1] \text{ w.p. 1}$$

$\lambda_1 =$ largest eigenvalue of $\underline{X}$

FACT: If $\underline{X}$ is symmetric, then

- ① all eigenvalues are real
- ② $\exists$ an orthonormal basis of eigenvectors

$$\lambda_1 = \sup_{\substack{\underline{v} \neq 0 \\ \|\underline{v}\|_2 = 1}} \underline{v}^T \underline{X} \underline{v}$$

If $\underline{X}$ is symmetric, $\{\underline{v}_1, \dots, \underline{v}_n\}$ eigenvectors (orthonormal)

$$\underline{X} = \sum_{i=1}^n \lambda_i \underline{v}_i \underline{v}_i^T$$

$$\begin{aligned} \underline{v}^T \underline{X} \underline{v} &= \sum_{i=1}^n \lambda_i \underline{v}^T \underline{v}_i \underline{v}_i^T \underline{v} \\ &= \sum_{i=1}^n \lambda_i \langle \underline{v}, \underline{v}_i \rangle^2 \end{aligned}$$

$$\begin{aligned} \underline{v} &= \sum_{i=1}^n \alpha_i \underline{v}_i & \|\underline{v}\|^2 &= \sum_{i=1}^n \alpha_i^2 \\ & & &= 1 \end{aligned}$$

$$\langle \underline{v}, \underline{v}_i \rangle = \langle \sum_j \alpha_j \underline{v}_j, \underline{v}_i \rangle = \alpha_i$$

$$\underline{v}^T \underline{X} \underline{v} = \sum_{i=1}^n \lambda_i \alpha_i^2, \quad \sum_{i=1}^n \alpha_i^2 = 1$$

$$\sup_{\underline{v}: \|\underline{v}\|_2=1} \underline{v}^T \underline{X} \underline{v} = \sup_{\underline{\alpha}: \sum_i \alpha_i^2=1} \sum_{i=1}^n \lambda_i \alpha_i^2$$

$$= \sup_{\substack{\beta_i \geq 0 \\ \sum_i \beta_i = 1}} \sum_{i=1}^n \beta_i \lambda_i$$

$$= \lambda_1$$

$$\lambda_1 = \sup_{\underline{v}: \|\underline{v}\|=1} \underline{v}^T \underline{X} \underline{v}$$

$$\lambda_1 = \underline{u}^T \underline{X} \underline{u}$$

Want to obtain an upper bound on variance of λ_1

$X_{ij} \in [-1, 1]$ ($X_{ij} : i \leq j$) are iid

$$\underline{X} = \begin{bmatrix} 1 & - & & 1 \\ & \ddots & & \\ & & \ddots & \\ 1 & - & - & 1 \end{bmatrix} \quad \lambda_1 = n$$

We make use of:

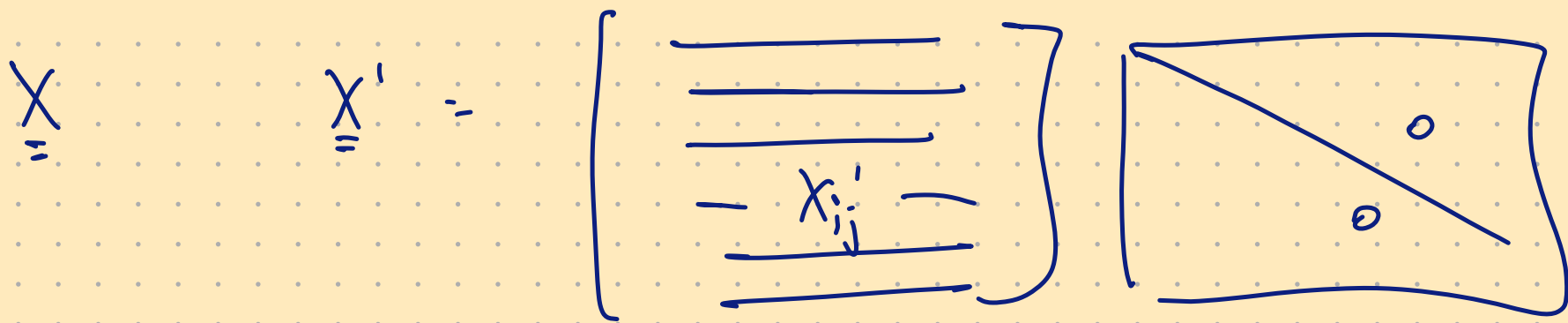
$$Z = g(X_1, \dots, X_k)$$

$$\text{Var}(Z) \leq \mathbb{E} \sum_{i=1}^k (Z - Z_i^*)^2$$

↓

obtained by replacing
 X_i with iid copy

$$\text{Var}(\lambda_1) \leq \mathbb{E} \sum_{i=1}^n \sum_{j \geq i} (\lambda_1 - \lambda_1^{(ij)})^2$$



$$(\lambda_1 - \lambda_{(ij)})_+ = \left(\sup_{\underline{v}: \|\underline{v}\|_2=1} \underline{v}^\top X \underline{v} - \sup_{\underline{v}: \|\underline{v}\|_2=1} \underline{v}^\top X' \underline{v} \right)_+$$

Suppose that $\underline{u} = \arg\max_{\|\underline{v}\|=1} \underline{v}^\top X \underline{v}$

$$\begin{aligned} (\lambda_1 - \lambda_{(ij)})_+ &\leq \left(\underline{u}^\top X \underline{u} - \sup_{\|\underline{v}\|=1} \underline{v}^\top X' \underline{v} \right)_+ \\ &\leq \left(\underline{u}^\top X \underline{u} - \underline{u}^\top X' \underline{u} \right)_+ \\ &= \left(\underline{u}^\top (X - X') \underline{u} \right)_+ \end{aligned}$$

$$= \left(\underline{u}^T \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \underline{u} \right) +$$

$$= \left(2u_i u_j (x_{ij} - x'_{ij}) \right) +$$

$$\begin{aligned} \left(\lambda_i - \lambda_i^{(i,j)} \right) + & \leq 4(u_i u_j) + \\ & \leq 4 |u_i u_j| \end{aligned}$$

$$\begin{aligned} \text{Var}(\lambda_i) &\leq \mathbb{E} \sum_{i \geq 1} \sum_{j \geq i} \left(\lambda_i - \lambda_i^{(i,j)} \right)_+^2 \leq \mathbb{E} \sum_{i=1}^n \sum_{j \geq i} 4^2 |u_i u_j|^2 \\ &\leq \mathbb{E} \left[16 \sum_{i=1}^n \sum_{j \geq i} u_i^2 u_j^2 \right] \end{aligned}$$

$$\leq 16 \mathbb{E} \sum_{i=1}^n \sum_{j=1}^n u_i^2 u_j^2$$

$$= 16 \mathbb{E} \left(\sum_{i=1}^n u_i^2 \right)^2 = 16$$

$\text{Var}(\lambda_1) \leq 16$ irrespective of n .

Self bounding functions

$f: \mathcal{X}^n \rightarrow \mathbb{R}$ is a self bounding fn if

$$(1) \quad 0 \leq f(x_1, \dots, x_n) \leq 1$$

$$(2) \quad \exists f_1, \dots, f_n$$
$$f_i: \mathcal{X}^{n-1} \rightarrow \mathbb{R} \quad \text{s.t.}$$

$$\sum_{i=1}^n f(x_1, \dots, x_n) = f_i(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$$
$$\leq f(x)$$

$$\Pr[f(X^n) \geq \mathbb{E}f(X^n) + \delta] \leq \exp\left[-\frac{\delta^2}{2\mathbb{E}f(X^n) + 2\delta/3}\right]$$

$$\Pr[f(X^n) \leq \mathbb{E}f(X^n) - \delta] \leq \exp\left[-\frac{\delta^2}{2\mathbb{E}f(X^n)}\right]$$



